MILITARY SPECIFICATION

SWITCHES, TOGGLE, HERMETICALLY SEALED, GENERAL SPECIFICATION FOR

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - This specification covers requirements for toggle-type hermetically sealed switches.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-up, Paperboard.
PPP-T-60	Tape, Pressure Sensitive Adhesive, Waterproof - for Packaging.
FED-STD-406	Test Method, Polycarbonate Materials.

MILITARY

MIL-I-10	Insulation, Ceramic
MIL-M-14	Molding Plastics and Molded Plastic Parts, Thermosetting
MIL-P-116	Preservation, Methods of
MIL-W-5086	Wiring Aircraft, 600 Volt Copper
MIL-C-5809	Circuit Breakers, Trip Free, Aircraft General Specification for
MIL-S-6872	Soldering Process, General Specification for
MIL-S-7742	Screw Threads, Std. Aero.
MIL-I-8500	Interchangeability and Replacement
MIL-F-15160	Fuses, Instrument, Power and Telephone
MIL-P-20693	Thermoplastic Material
MIL-C-45662	Calibration System Requirements
MIL-I-81023	Inductor, 28 v-dc, Laboratory Test, General Specification for
MIL-T-7928	Terminals, Lug ans Splice, Crimp-Style, Copper

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures & Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSC 5930

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MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-202	Test Methods for Electronic and Elec. Component Parts.
MIL-STD-456	Electronic Parts, Date and Source coding for.
MS25017	Circuit Breaker, Trip Free, Push Pull, 5 thru 50 AMP. Type 1
MS33586	Metals, definition of dissimilar

FEDERAL

FED. STD. 406	Plastics, Methods of Testing
ZZ-R-765	Rubber, Silicone, Low and High Temperature and Tear Resistant

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

3. REQUIREMENTS

3.1 Detail Specifications - The individual switch requirements shall be as specified herein and in accordance with the applicable detail specifications. In the event of any conflict between this specification and the detail specifications, the latter shall govern.

3.2 Qualification - Switches furnished under this specification, and covered by detail specifications shall be a product which has been tested, passed the qualification tests specified in Paragraph 4.0, and has been listed on or approved for listing on the applicable qualified products list.

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3.3 Material - Material shall be as specified herein. However, when a definite material is not specified, a suitable material shall be used which will enable the switches to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product.

3.3.1 Metals - All metal parts, other than current-carrying parts, shall be of corrosion resistant material or shall be suitably protected to resist corrosion.

3.3.1.1 Ferrous Material - Ferrous metal shall not be used for current-carrying parts except for feed-through terminals.

3.3.1.2 Dissimilar Metals - When dissimilar metals are used in intimate contact with each other, protection against electrolysis and corrosion shall be provided. The use of dissimilar metals in contact, which tend toward active electrolytic corrosion (particularly brass, copper or steel used in contact with aluminum or its alloys) is not acceptable. However, metal plating or metal spraying of dissimilar base metals to provide similar or suitable abutting surfaces is permitted. The use of dissimilar metals separated by a suitable insulating material is also permitted. Dissimilar metals within hermetically sealed enclosure will be permitted. Dissimilar metals are defined in MS 33586.

3.3.2 Insulation -

3.3.2.1 Ceramic - Ceramic insulation shall be grade L422 or higher, in accordance with MIL-I-10.

3.3.2.2 Plastic - Unless otherwise specified (see 3.1) thermosetting plastics shall be in accordance with MIL-M-14; however, cotton or wood flour filled materials shall not be used. Thermoplastics shall not be used for the switch case or cover. When used for non-case material, thermoplastic material shall be in accordance with MIL-P-20693 (Type I), or shall be of a high temperature, polycarbonate material which is self-extinguishing when tested in accordance with Method 2021 of FED-STD-406.

3.3.3 Elastomer Seals - The material for elastomer seals shall be silicone rubber in accordance with ZZ-R-765.

3.3.4 Glass - Glass of high purity only shall be used in glass-to-metal seals.

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3.4 Design and Construction - Switches shall be so constructed as to insure proper operation when mounted in any position. Switches shall be of the design, construction and physical dimensions specified in the detail specifications. (see Paragraph 3.1)

3.4.1 Interchangeability - All switches having the same part number shall be directly and completely interchangeable with each other, in accordance with MIL-I-8500.

3.4.1.1 Size and Weight - Size and weight shall be as shown in the applicable detail specifications.

3.4.1.2 Actuating Lever - The actuating lever shall have a non-glare lusterless finish and shall be insulated from current carrying parts.

3.4.2 Actuating Means - The actuating mechanism shall be as shown in the detail specifications.

3.4.2.1 Non rotating and grounded to bushing.

3.4.2.2 Operating force direction shall be perpendicular to the direction of toggle travel.

3.4.2.3 Point at which operating force shall be applied to the toggle shall be in accordance with the detail specifications.

3.4.2.4 For lever lock type, release of toggle shall be accomplished by a force applied axially to the lever.

3.4.2.5 Operating arc requirements shall be in accordance with the detail specifications.

3.4.2.6 Operating force requirements shall be as shown in detail specifications.

3.4.3 Mounting Means and Hardware - Switches shall be of the single hole mounting type. Assemblies shall include a no-turn device. Mounting hardware shall be provided as specified in the detail specifications.

3.4.4 Terminals - Terminals and associated hardware shall be as specified in the detail specifications. Lockwashers, as required shall be captive to the screw.

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3.4.4.1 Terminal Identification - Switches shall have a circuit diagram and terminals shall be identified to indicate contact arrangement in accordance with detail specifications.

3.4.5 Screw Threads - Screw threads on removable or replaceable threaded parts shall be as specified in MIL-S-7742. Threading of non-metallic parts shall not be permitted.

3.4.6 Soldering - Solder joints shall be processed according to MIL-S-6872.

3.4.6.1 Corrosive fluxes shall not be used. When stainless steel is used on hermetically sealed switches, corrosive flux may be used on sub-assemblies providing cleaning and neutralizing is accomplished prior to the final assembly sealing takes place. Solder shall not be used to obtain mechanical strength.

3.5 Marking of switches - Switches shall be permanently and legibly marked in accordance with MIL-STD-130. The following information shall be included:

Nomenclature	Circuit Diagram
Mfr. Part Number	Voltage and current rating
Military Part Number	Mfg. Date per MIL-STD-456.

3.6 Workmanship - Switches shall be processed in such a manner as to be uniform in quality, free from cracked or displaced parts, sharp edges, burrs and any other defects which could affect life, serviceability and performance.

3.7 Performance Requirements

3.7.1 Operating Characteristics - Switches shall be tested for operating characteristics in accordance with the requirements of the detail specifications (See Paragraph 4.7.2).

3.7.1.1 Coincidence of Operate and Release Points (Multi-pole switches only). Switches tested as specified in (Paragraph 4.7.2.1) all relate to the circuit contacts as specified in the applicable detail specifications and shall have made contact within the limits specified in the detail specifications.

3.7.2 Strength of Terminals - When switches are tested as specified in Paragraph 4.7.3 there shall be no evidence of short circuiting, breakage, loosening, rotation of terminals or damage to switch body.

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3.7.3 Strength of Toggle - When switches are tested as specified in Paragraph 4.7.4, there shall be no electrical or mechanical damage during or after the test and operating characteristics shall be as specified in Paragraph 4.7.2.

3.7.4 Strength of Mounting Bushing - Switches tested as specified in Paragraph 4.7.4.5 shall show no evidence of damage to the switch assembly, loosening or twisting of the bushing relative to the switch assembly and shall maintain operating characteristics as specified in Paragraph 4.7.2.

3.8 Environmental and Electrical Requirements

3.8.1 Thermal Shock - When switches are tested as specified in paragraph 4.7.5, there shall be no electrical or mechanical damage, or loosening of rivets or other fastening devices.

3.8.2 Vibration - When switches are tested as specified in Paragraph 4.7.6, there shall be no evidence of broken, loose deformed or displaced parts. There shall be no evidence of closed contacts opening or open contacts closing in excess of 10 microseconds.

3.8.3 Shock - When switches are tested as specified in Paragraph 4.7.7, there shall be no evidence of broken, deformed, loose or displaced parts. There shall be no mechanical or electrical damage and no opening of closed contacts or closing of open contacts in excess of 5 milliseconds.

3.8.4 Acceleration - When switches are tested as specified in Paragraph 4.7.8, there shall be no evidence of closed contacts opening or open contacts closing in excess of 5 milliseconds. There shall be no evidence of electrical or mechanical damage.

3.8.5 Moisture resistance - Switches shall be tested as specified in Paragraph 4.7.9. Immediately after conclusion of the test and while the switches are still in the humidity chamber, the insulation resistance shall be as follows

Ceramic - Greater than 50 megohms
Plastic - Greater than 10 megohms

At the end of the drying period, the insulation resistance shall be as specified in Paragraph 3.8.6. At the conclusion of the test there shall be no

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evidence of corrosion, breaking, cracking, spalling or loosening of terminals and mounting hardware which would be readily removable.

3.8.6 Insulation Resistance - When measured as specified in Paragraph 4.7.10, the insulation resistance shall be 1,000 megohms minimum.

3.8.7 Dielectric Withstanding Voltage - When switches are tested as specified in Paragraph 4.7.11 and 4.7.12, there shall be no arcing, flashover, leakage or breakdown in excess of .5 milliamperes.

3.8.8 Salt Spray - When switches are tested as specified in Paragraph 4.7.13, there shall be no evidence of excessive corrosion. Excessive corrosion is defined as that which interferes with the electrical and/or mechanical performance. There shall be no warping, cracking or other damage to the switch assembly. After the test the mounting hardware shall be readily removable.

3.8.9 Explosion - When switches are tested as specified in Paragraph 4.7.23, there shall be no explosion within the test chamber whether or not an explosion occurs within the switch.

3.8.10 Sand and Dust - When switches are tested as specified in Paragraph 4.7.14, there shall be no evidence of physical damage and the switches shall be mechanically and electrically operative at the conclusion of the test. The evidence shall be based on their ability to control a pilot lamp circuit.

3.8.11 Mechanical Endurance - When switches are tested as specified in Paragraph 4.7.15, there shall be no evidence of mechanical damage and contact resistance shall be as specified in Paragraph 3.8.12.

3.8.12 Contact Resistance - When measured as specified in Paragraph 4.7.16, the contact resistance shall not exceed the values below with contacts carrying the specified test current.

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	Open Circuit D.C. Volts	Test Current Amperes	Contact Resistance Ohms
Before Testing			
	28	10.0	0.015
	6	0.100	0.025
After All Testing			
	28	10.0	0.025
	6	0.100	0.050

3.8.13 Low Temperature Operation - When switches are tested as specified in Paragraph 4.7.17, the operating characteristics shall be in accordance with Paragraph 3.7.1 and the contact resistance shall be in accordance with Paragraph 3.8.12 subsequent to the test.

3.8.14 Icing Operation - When switches are tested as specified in Paragraph 4.7.18, the switches shall actuate immediately without damage when the force is applied to the actuating means at 25°F and -65°F. Repeated application of the icing operation test shall not result in a mechanical or electrical failure of the switch. After the test, the switches shall be tested for proper operation for 50 cycles at rated resistive load.

3.8.15 Ozone - When switches are tested as specified in Paragraph 4.7.19, there shall be no cracking of materials or other damage which will adversely affect subsequent performance of switch.

3.8.16 Sealing - When switches are tested as specified in Paragraph 4.7.20, the leakage rate shall not exceed the limits specified.

3.8.17 Short Circuit - When switches are tested as specified in Paragraph 4.7.21, there shall be no welding or sticking of contacts or damage to the switch. The switches shall be mechanically and electrically operative at the end of the test.

3.8.18 Overload Cycling - When switches are tested as specified in Paragraph 4.7.22, there shall be no mechanical or electrical failure. The contact resistance shall be as specified in Paragraph 3.8.12.

3.8.19 Electrical Endurance - When switches are tested as specified in 4.7.24, no contact shall fail to open.

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or close its individual circuit in proper sequence. After the test, the switches shall be electrically and mechanically operative. The contact resistance shall be as specified in Paragraph 3.8.12.

3.8.20 Altitude - The altitude rating shall be (80,000 feet) .820 inch mercury absolute.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test Equipment and Facilities - Test equipment and facilities shall be of sufficient accuracy, quality and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government. Calibration of the standards which control the accuracy of the inspection equipment shall comply with the requirements of MIL-C-45662.

4.2 Classification of Inspection - The examination and testing of switches shall be classified as follows:

(a) Component materials inspection (See 4.3)
(b) Qualification Inspection (See 4.5)

(c) Quality Conformance Inspection. (See 4.6)

4.3 Component Materials Inspection - Component materials inspection shall consist of verification that the component materials listed in Table I, used in fabricating the switches are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

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TABLE I

COMPONENT MATERIALS INSPECTION

Component Material	Required Paragraph	Applicable Specifications
Ceramic	3.3.2.1	Grade L422 or Higher MIL-I-10
Plastic	3.3.2.2	MIL-M-14 MIL-P-20693(Type I) FED-STD-406 Method 2021
Elastomer (Rubber) (Silicone)	3.3.3	

4.4 Inspection Conditions - Unless otherwise specified, herein, all inspections shall be made at room ambient temperature, relative humidity and pressure.

4.5 Qualification Inspection - Qualification inspection shall be performed at a laboratory satisfactory to the Government.

4.5.1 Sample - For qualification inspection, the sample size for each test shall be as specified in Table II. Total number of samples shall be twenty-four (24).

4.5.1.1 Single Submission - A sample consisting of the required number of specimens of each switch shall be submitted. Qualification shall be restricted to the type submitted.

4.5.1.2 Failures - Failure of any switch to comply with the applicable requirements shall be cause for rejection.

4.5.1.3 Test Data - The Contractor will prepare test data sheets in accordance with paragraph 4.2 for each classification of test. Every switch test shall have a test data sheet indicating quantitative data at the time of testing. Test data sheets shall be prepared in duplicate and the original given to the Government Inspection Activity responsible for acceptance.

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TABLE II Page 1 of 2

QUALIFICATION TEST SEQUENCE

Examination or Test	Requirement Paragraph	Test Paragraph	Switch Sample Number and Sequence																							
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Examination of Product (24)	3.1 3.3 3.4 3.5 3.6	4.7.1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Seal Test Hermetic (24)	3.8.16	4.7.20.1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
Seal Test Gross (24)	3.8.16	4.7.20.2	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
Operating Characteristics	3.7	4.7.2	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
Contact Resistance	3.8.12	4.7.16	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5	5
Strength of Terminals	3.7.2	4.7.3	6	6																						
Strength of Toggle	3.7.3	4.7.4			6	6																				
Strength of Mounting Bushing	3.7.4	4.7.4.5					6	6															6	6		
Thermal Shock	3.8.1	4.7.5	7	7																						
Vibration	3.8.2	4.7.6			7	7	7	7																	9	9
Shock	3.8.3	4.7.7					8	8															7	7		
Acceleration	3.8.4	4.7.8					9	9																		
Salt Spray	3.8.8	4.7.13					10	10																		
Moisture Resistance	3.8.5	4.7.9			8	8																				
Insulation Resistance	3.8.6	4.7.10	8	8					6	6																
Mechanical Endurance	3.8.11	4.7.15	9	9																						
Low Temperature	3.8.13	4.7.17							7	7																
Icing Operation	3.8.14	4.7.18								8	8															
Ozone	3.8.15	4.7.19										6	6											8	8	
Sand and Dust	3.8.10	4.7.14																								

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TABLE II page 2 of 2

QUALIFICATION TEST SEQUENCE

Examination or Test	Requirement Paragraph	Test Paragraph	Switch Sample Number and Sequency																						
			1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Inductive Load, DC	3.8.19	4.7.25.2																6								
Lamp Load, DC	3.8.19	4.7.25.3																	6							
Resistive Load, AC	3.8.19	4.7.25.4																		6						
Inductive Load, AC	3.8.19	4.7.25.5																			6					
Lamp Load, AC	3.8.19	4.7.25.6																				6				
Dielectric Strength	3.8.7	4.7.12 4.7.11			9	9																	9	9		
Insulation Resistance	3.8.6	4.7.10			10	10	11		11																	
Operating Characteristics	3.7	4.7.2									7	7													9	9
Contact Resistance	3.8.12	4.7.16	10	10																			10	10	10	10
Seal, Gross	3.8.16	4.7.20.2	11	11		12	12																11	11	11	11
Seal, Hermetic	3.8.16	4.7.20.1	12	12		13	13																			
Explosion	3.8.9	4.7.23	13	13																						

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4.5.1.4 Certification of Material - When submitting samples for qualification, the manufacturer shall submit certification in duplicate, that the materials used in switch components are in accordance with the applicable specification requirements.

4.5.1.5 Description of Items - The manufacturer shall submit his part number for each switch for which qualification is sought.

4.6 Quality Conformance Inspection

4.6.1 Inspection of Product for Delivery - Inspection of product for delivery shall consist of Group "A" inspection.

4.6.1.1 Group "A" Inspection - Group "A" inspection shall consist of the examination and tests specified in Table III, in the order shown at room ambient conditions. Statistical sampling and inspection shall be in accordance with MIL-STD-105. The acceptable quality levels (AQL) shall be as defined in MIL-STD-105.

4.6.1.2 Inspection Lot - All switches of the same part number offered for delivery at the same time shall be considered as one lot for the purposes of sampling and inspection.

4.6.1.3 Rejected Lots - If an inspection lot is rejected, the supplier shall withdraw the lot and may then rework to correct defects, or screen out the defective units. Such lots shall be kept separate from all other lots and shall be clearly identified as lots to be reinspected. Reworked lots shall be reinspected using tightened inspection plans according to MIL-STD-105.

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TABLE III

	Requirement	Method
GROUP A (100% Inspection)		
Continuity Check	3.7.1	4.7.2
Seal Test	3.8.16	4.7.20.1, 4.7.20.2
Dielectric Strength	3.8.7	4.7.11, 4.7.12
Insulation Resistance	3.8.6	4.7.10
Contact Voltage Drop	3.8.12	4.7.16
Coincidence of Operation	3.7.2	4.7.2.1
Examination of Product		
* GROUP B (MAJOR)		
Operating Force	3.4.2	4.7.2
Terminal Strength	3.7.2	4.7.3
Toggle Strength	3.7.3	4.7.4
Strength of Mounting Bushing	3.7.4	4.7.4

* Sampling and Acceptable Quality Level - For Group B tests, sampling shall be in accordance with MIL-STD-105, Inspection Level II, AQL 1.5.

4.6.2 Retention of Qualification - In order to retain qualification, at yearly intervals the manufacturer shall forward a summary of the results of Group "B" tests performed during the year to the activity responsible for qualification approval via the Government Inspector. Retention of qualification approval will be based on the "C" test results. Failure to submit the summary will result in loss of qualification for the product involved. When production has been suspended for a year or more, the activity responsible for qualification shall be notified, and group "B" tests shall be performed on the next lot produced.

4.6.2.1 Group "C" Inspection - Group "C" Inspection shall consist of the tests specified in Table IV in the order shown.

4.6.2.2 Sampling Plan - Switches shall be selected by the Government Inspector once each year following initial product qualification and subjected to Group "C" inspection. All tests shall be performed on sample units that have passed Group "A" and "B" inspection.

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TABLE IV GROUP "C" TEST
REQUALIFICATION TEST SEQUENCE

Examination or Test	Requirement Paragraph	Test Paragraph	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
Examination of Product	3.1 3.3 3.4 3.5 3.6	4.7.1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
Seal, Hermetic	3.8.16	4.7.20.1	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2	2
Seal, Gross	3.8.16	4.7.20.2	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3	3
Operating Characteristics	3.7	4.7.2	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4	4
Strength of Terminals	3.7.2	4.7.3	5	5																						
Strength of Toggle	3.7.3	4.7.4			5	5																				
Strength of Mounting Bushing	3.7.4	4.7.4.5					5	5																		
Mechanical Endurance	3.8.11	4.7.15	6	6																	5	5				
Overload Cycling	3.8.18	4.7.22			6	6																	5	5		
Electrical Endurance	3.8.19	4.7.24																								
Resistive Load DC	3.8.19	4.7.25					6																			
Inductive Load DC	3.8.19	4.7.25.1						6																		
Lamp Load, DC	3.8.19	4.7.25.2							5																	
Resistive Load AC	3.8.19	4.7.25.3								5																
Inductive Load AC	3.8.19	4.7.25.4									5															
Lamp Load, AC	3.8.19	4.7.25.5										5														
Dielectric Strength	3.8.7	4.7.11 4.7.12											5 6	5 6											5 6	5 6
Vibration	3.8.2	4.7.6											7	7											7	7
Shock	3.8.3	4.7.7											8	8											8	8
Acceleration	3.8.4	4.7.8													5	5										
Salt Spray	3.8.8	4.7.13															5	5								
Moisture Resistance	3.8.5	4.7.9																	5	5						
Sand and Dust	3.8.10	4.7.14																								
Examination of Product	3.1 3.3 3.4 3.5 3.6	4.7.1	7	7									9 10	9 10	6	6	6	6	6	6	6	6	6	6	9 10	9 10
Seal, Hermetic	3.8.16	4.7.20.1	8	8									11	11											11	11
Seal, Gross	3.8.16	4.7.20.2	8	8																		7	7			
Explosion	3.8.9	4.7.23																								

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4.6.2.3 Disposition of Sample Units - Sample units subjected to Group "C" inspection shall not be delivered on the contract or order.

4.7 Examination and Test Procedures

4.7.1 Examination of Product - Switches shall be examined to verify that materials, design, construction, dimensions, marking and workmanship are in accordance with this specification (See Para. 3.1 thru 3.6).

4.7.2 Operating Characteristics - (See Para 3.7.1) Switch shall be tested for operating characteristics in accordance with the requirements of the detail specifications and Table III Groups "A" and "B" inspection.

4.7.2.1 Coincidence of Operating and Release Points - (See Para 3.7.11) Coincidence of operating or release points of all poles of a multipole switch shall be determined with suitable indicating circuits. The switch shall be rigidly held in a suitable fixture, allowing the actuator to be moved in its intended manner. The actuator shall be advanced and returned slowly and uniformly with no external vibratory influence while passing through the operating and release points, at a rate not exceeding 0.002" or one (1) degree persecond. The circuit must remain in its respective closed or open position for a minimum of 4° as measured from a rest position or from full travel in a momentary position. On switches with more than one rest position, once the circuit is broken, the circuit shall not remake when actuating force is removed and the toggle must travel to the adjacent rest position. If force is removed prior to breaking or making the circuit, the toggle must return to its respective rest position.

4.7.3 Strength of Terminals - (See Para 3.7.2) Switches shall be mounted by their normal mounting means.

4.7.3.1 Screw Terminals - Screw terminals shall be subjected to the tests specified in Paragraphs 4.7.3.1.1 and 4.7.3.1.2.

4.7.3.1.1 Pull Test - The terminals shall be subjected to a pull of the applicable static force specified in Table V in a direction along the axis of the terminal screw, in a direction perpendicular to the axis of the screw terminal and in the direction most likely to cause failure. The force shall be applied for one minute in each of the directions. The switches shall be examined for evidence of short circuiting breakage, loosening of terminals or damage to the body of the switch. No terminal shall be tested in more than one (1) direction.

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4.7.3.1.2 Torque Test - Terminals of the switches shall have a torque test as specified in Table V applied to the head of the terminal screw in a plane parallel to the face of the terminals, clockwise and then counterclockwise without failure of any type. The switch shall be examined for evidence of short circuiting, breakage, loosening of terminals or damage to the body of the switch.

TABLE V

Thread Size	Static Pull Force (Lbs)	Torque Inch-Pounds
6-32	30	15

4.7.3.1.3 Strength of Solder Type Pierced Terminals - Solder type pierced terminals shall be tested as follows:

4.7.3.1.4 Pull Test - The terminals shall be subjected to a static pull test of 10 (ten) pounds. Force shall be applied parallel to the center line of the actuating lever.

4.7.3.1.5 Torque Test - A torque test of 5 inch pounds shall be applied to the terminals for a period of 1 minute. This test shall be made in clockwise and counterclockwise directions.

4.7.3.1.6 Examination After Testing - Switches shall be examined for evidence of short circuiting, breakage, loosening of terminals, damage to the switch body and loss of hermetic seal.

4.7.4 Strength of Toggle -(See 3.7.3) With the switch carrying rated resistive load at the maximum voltage specified in the detail specification and with the switch mounted in its intended manner a static load of twenty-five (25) pounds shall be gradually applied to the tip of the lever and maintained for a period of one minute under each of the

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following conditions: (Contacts to be monitored for proper operation during test).

4.7.4.1 Perpendicular to the lever axis and parallel to the lever travel at each extreme position of the lever.

4.7.4.2 Perpendicular to the lever axis and perpendicular to the line of travel throughout the entire range of the lever travel.

4.7.4.3 Co-axial with the lever axis toward the lever pivot throughout the entire range of the lever travel.

4.7.4.4 Co-axial with the lever axis away from the lever pivot throughout the entire range of the lever travel.

4.7.4.5 Strength of Mounting Bushing - (See 3.7.4) Switches shall be mounted on a metal panel in their normal mounting means with the hardware specified (see 3.4.3). A torque of 15 inch-pounds shall be applied to the mounting nut clockwise then counterclockwise to loosen the nut. Test the no turn device by subjecting the mounted switch assembly to a torque of 5 inch-pounds clockwise then counterclockwise.

4.7.5 Thermal Shock - (See 3.8.1) Switches shall be tested in accordance with method 107 of MIL-STD-202. The test conditions shall be as specified in Table VI.

TABLE VI

TEST CONDITIONS

Step	Temperature ° (F)	Time (Minutes)
1	$-65\ ^{+0}_{-5}$	30 minimum
2	$77\ ^{+15}_{-10}$	5 maximum
3	$275\ ^{+5}_{-0}$	30 minimum
4	$77\ ^{+15}_{-10}$	5 maximum

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4.7.6 Vibration - (See Para 3.8.2) Switches shall be tested in accordance with Method 204 of MIL-STD-202, Test Condition "D" as defined in figure I except using the vibration schedule as shown in Table VII.

4.7.6.1 Vibration Input - The applied double amplitude or vibratory acceleration shall be in accordance with Appendix "A". All frequency measurements shall be accurate within ±5 per cent. All amplitude or acceleration measurements shall be accurate within ±10 per cent.

4.7.6.2 Mounting - Switches shall be rigidly mounted by their normal mounting means on a rigid metal panel. The mounting fixture shall be free of resonances over the test frequency range.

4.7.6.3 Contact Monitoring - All open and closed circuits must be monitored. The test time shall be divided so that each switch has approximately equal time in each of its positions, except that, switches with momentary position(s) need not be tested with the switch in the momentary position(s). Open circuits may be connected in parallel and monitored for closing and closed circuits may be connected in series and monitored for opening. In the event of an indication of a failure, the test shall be continued to determine which switch is defective. If one or more contacts fail, the switch shall be considered a failure.

4.7.6.4 Monitoring Test Measurements - Switch contact stability shall be continuously monitored during vibration by an acceptable thyratron type test circuit on other comparable means capable of detecting opening of closed contacts or closing of open contacts, in excess of 10 microseconds duration. In the event of indication of contact opening or closing greater than allowed, the vibration cycle shall be continued long enough to monitor individual switch contacts to determine which switch is defective.

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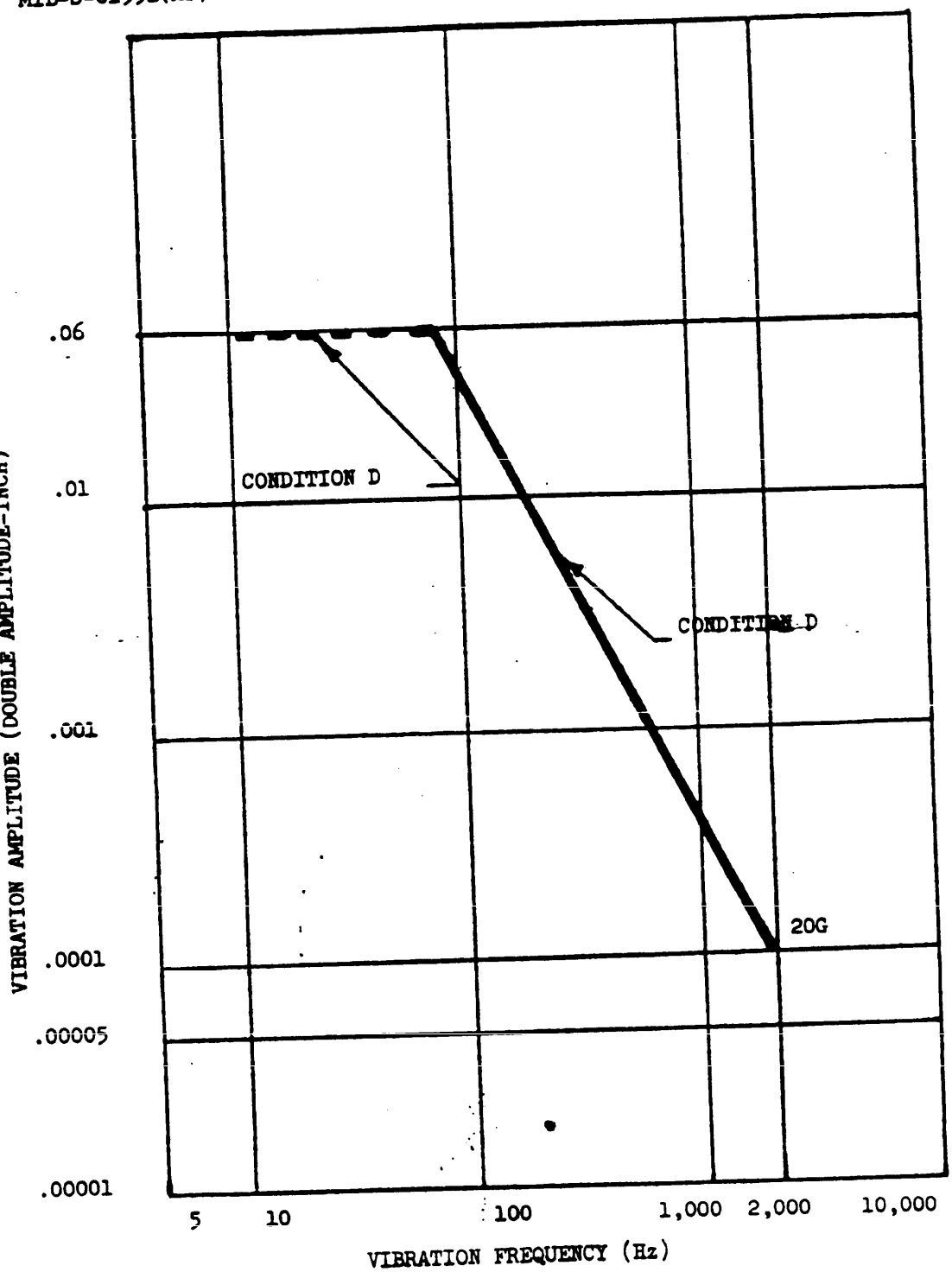


FIGURE I

VIBRATION-TEST CURVES

METHOD 204 of MIL-STD-202
CONDITION "D"

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4.7.6.5 Resonance and Cycling Endurance Test

4.7.6.5.1 $-65°F\ ^{+0}_{-5}$ Two (2) 15 minute vibration cycles.

4.7.6.5.2 $+160°F\ ^{+10}_{-0}$ Two (2) 15 minute vibration cycles.

4.7.6.5.3 Room ambient conditions - in accordance with vibration test schedule, Table VII.

TABLE VII

Number of Resonances	0	1	2	3
Total Vibration time at Resonance	0	1/2 hr	1 hr	1-1/2 hr
Total Cycling Time (15 min per cycle)	2 hr	1-1/2 hr	1 hr	1/2 hr

4.7.7 Shock - (See Para 3.8.3) Switches shall be tested in accordance with Method 213 of MIL-STD-202, Test Condition "H".

4.7.7.1 Mounting Means - Switches shall be mounted on a rigid metal panel by their normal mounting means.

4.7.8 Acceleration - (See Para 3.8.4) Switches shall be subjected to the acceleration test of MIL-STD-202, Method 212, test condition "C" at a force of 20 g's.

4.7.8.1 Contact Monitoring - The switch contacts shall be monitored in the manner described in Para 4.7.6.3 and 4.7.6.4, duration of contact opening from closed position or closing from open position shall not exceed 500 microseconds.

4.7.9 Moisture Resistance - (See Para 3.8.5) Switches shall be tested in accordance with Method 106 of MIL-STD-202. The following details and exceptions shall apply.

4.7.9.1 Mounting - Mount by normal mounting means on the upper side of a corrosion resistant metal panel positioned 15 degrees from vertical.

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4.7.9.2 Steps 7A and 7B - not applicable.

4.7.9.3 Polarization - During steps 1 to 6 inclusive, a polarization voltage of 100v-dc shall be applied between the metal panel and the terminals of the switch, with all terminals connected together. The negative polarity shall be applied to the metal panel.

4.7.9.4 Load voltage not applicable.

4.7.9.5 Final measurements - Immediately after conclusion of the test and while switches are still in the humidity chamber, the insulation resistance shall be measured as specified in Para 4.7.10.

4.7.10 Insulation Resistance - (See Para 3.8.5, 3.8.6) The insulation resistance test shall be in accordance with Method 302 of MIL-STD-202. The following details shall apply.

4.7.10.1 Test condition - "B" (~~500~~ volts) minimum resistance 1000 megohms.

4.7.10.2 Points of measurement.

4.7.10.3 Between each terminal and ground.

4.7.10.4 Between all adjacent contact terminals of different poles.

4.7.10.5 Between all unconnected terminals of the same pole.

4.7.11 Dielectric Withstanding Voltage - (See Para 3.8.7)

4.7.11.1 At Atmospheric Pressure - Switches shall be tested in accordance with Method 301 of MIL-STD-202. The following details and exceptions shall apply.

4.7.11.2 Test Voltage - 1000 volts root mean square (VRMS) maximum leakage .5MA.

4.7.11.3 Duration of application - One (1) minute.

4.7.11.4 Points of Application - Between each terminal and ground. Between all adjacent contact terminals of different poles. Between all unconnected terminals of the same pole.

4.7.12 At Reduced Barometric Pressure - Switches shall be

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4.7.12 (cont.)

tested as specified in Para 4.7.11 and in accordance with Method 105 of MIL-STD-202. The following details and exceptions shall apply.

4.7.12.1 Test Voltage - 400 VRMS maximum leakage .5MA.

4.7.12.2 Test Condition Altitude - (80,000 feet) .820 inch Mercury absolute.

4.7.13 Salt Spray (See Para 3.8.8) Switches with associated mounting hardware, shall be tested in accordance with Method 101 of MIL-STD-202. The following details and exceptions shall apply.

4.7.13.1 Test Condition - letter "A" (96 hours).

4.7.13.2 Final Measurements - At the completion of the test, switches shall be subjected to a 6 hour drying period in a forced draft oven at a temperature of approximately 135°F. The switch shall then be subjected to 10 cycles of operation of making and breaking the rated resistive current at the lowest dc voltage specified in the detail specifications.

4.7.14 Sand and Dust - (See Para 3.8.10) Switches shall be tested in accordance with Method 110, Test Condition "B", of MIL-STD-202.

4.7.14.1 Operating Cycles - During the test, the switches shall be mechanically operated for 2500 cycles. The cycling rate shall not exceed 60 cycles per minute. A cycle is defined in Para 4.7.18.4.

4.7.15 Mechanical Endurance - (See Para. 3.8.11) One switch shall be tested for 50,000 cycles of operation at room ambient temperature. One switch shall be tested for 15,000 cycles of operation at $-65°F^{+0}_{-5}$ followed by 15,000 cycles of operation at $160°F \pm 10$. The cycling rate shall be 10 to 18 cycles per minute. A cycle is defined in para. 4.7.18.4. (At the manufacturer's option, a faster rate not to exceed 60 cycles per minute may be used.)

4.7.16 Contact Resistance - (See Para 3.8.12) Switches shall be tested in accordance with Method 307 of MIL-STD-202. The following details and exceptions shall apply.

4.7.16.1 Measurement shall be made between terminals of contacts of the same pole forming a switching circuit. Measurements shall be made for all poles in a switch.

4.7.16.2 Test Current and Voltage - Each switching

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4.7.16.2 (cont.)

circuit of the same pole shall be connected to loads in accordance with the following schedule and sequence.

Test	Pole	Test Current (Amperes)	Open Circuit Test Voltage
A	All poles	0.100	6.0v-dc
B	All poles	10.0	28.0±2v-dc

4.7.16.3 Resistance Values - Contact resistance in excess of the values specified in Para 3.8.12 shall constitute failure.

4.7.16.4 Test Cycles - Three cycles per pole prior to measurement. Three test cycles per pole. Record measurement for each of three test cycles.

4.7.17 Low Temperature Operation - (See ~~3.8.13~~) Switches shall be subjected to -65°F +0 -5 for a period of time sufficient to reach temperature stabilization. When the temperature has been stabilized, the units shall be subjected to contact resistance test of Para 4.7.16 and operating characteristics of Para 4.7.2.

4.7.18 Icing Operation - (See Para 3.8.14) Switches shall be carefully cleaned to remove grease or other foreign matter from surfaces that will be involved in the tests.

4.7.18.1 Ice Coating Procedure - Clear glaze-ice coating shall be formed over the switch alternately dipping it in distilled tap water and a cold chamber held at approximately 25°F. The ice coating shall be not less than 1/8 inch thick. Following the formation of the minimum ice coating, an aging period of not less than 20 minutes shall occur with the iced switch held at approximately 25°F.

4.7.18.2 Normal Position Test - The ice coating shall be as specified in Para 4.7.18.1 with the toggle in the normal position. The toggles shall be made to actuate by a force not exceeding 25 pounds. The switch shall operate immediately without damage. If the actuated position is a momentary (spring loaded) position, the switch toggle shall release immediately from the actuated position when the actuating force is removed. The switch shall be made to

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4.7.18.2 (cont)

actuate three additional times to assure proper operation.

4.7.18.3 Actuated Position Test - The ice coating shall be applied as specified in Para 4.7.18.1 with the toggle in each of the actuated positions. For either momentary or non-momentary actuated positions, a force not exceeding 25 pounds may be applied to break the ice coating and release the switch on the initial operation. For all subsequent operations of momentary actuated positions, the switch toggle shall release immediately from the actuated position when the force is removed. The switch shall be made to operate three additional times in each actuated position to assure proper operation.

4.7.18.4 Cycle of Operation - A cycle of operation shall be the movement of the actuating means through the entire range of its travel causing the switch contacts to change from one position to another and then return to their original position. Each stroke of the actuating means shall include a full range of travel from free position to full overtravel position and return to the fully released position.

4.7.18.5 Repeat the ice coating and aging operation of Para 4.7.18.1 immediately following the aging period, place the switch assemblies in a cold chamber at -65°F and leave until the temperature of the assemblies have stabilized, then repeat the tests of para 4.7.18.2 and 4.7.18.3

4.7.19 Ozone - (See para 3.8.15) Switches shall be placed in an enclosure at room temperature and subjected to an atmosphere having an ozone concentration of from 0.010 to 0.015 per cent by volume for a period of 2 hours. At the end of this period, the sample shall be removed and examined for signs of ozone deterioration. Switches shall be fitted with fixtures that apply the forces required of the tests. All tests shall use a suitable indicating device that will determine that the contacts operate properly.

4.7.20 Sealing Test - (See Para 3.8.16)

4.7.20.1 Hermetic Seal - Switches shall be backfilled prior to final weld with a 90% N 10% He gas. Switches shall be tested in accordance

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4.7.20.1 (Cont)

with Method 112A of MIL-STD-202. The following details shall apply.

(a) Test Condition "C"
(b) Leakage rate sensitivity 1 x 10^{-8} ATM cc/sec
(c) Procedures I - II - III or IV

4.7.20.2 Watertight Seal Test - (Gross Leak) Switches shall be immersed in a container of water containing approximately 1% aerosol (wetting agent) and shall then be placed in a vacuum chamber. The chamber shall then be evacuated to an absolute pressure of 1.3 inches of mercury, this pressure shall be maintained for a period of 1 minute, or until air bubbles cease to be given off by the water. The absolute pressure shall then be increased to 2.5 inches of mercury and this pressure shall be maintained for a period of 2 minutes. During the 2 minute period, the switches shall be observed for evidence of a continuous stream of bubbles. Any bubbles coming from within the switches shall be considered leakage. Bubbles which are the result of entrapped air on the exterior of the switches shall not be considered as an indication of leakage.

4.7.21 Short Circuit - (See Para 3.8.17) Switches shall be inserted in a circuit, calibrated to supply current equal to 60 times the rated resistive load at the lowest rated dc voltage specified in the detail specifications. The switch shall be connected in series, by one foot length of wire, to a thermal type circuit breaker in accordance with Figure II. Refer to Table VIII for wire size and circuit breaker designation.

4.7.21.1 Short Circuit Procedure - Calibration shall be made with the calibrate switch (S2) in the calibrate position.

4.7.21.2 With the circuit breaker and the source switch (S1) closed and the calibrate switch (S2) in the test position, the circuit shall be closed by the switch under test. The switch under test shall remain closed until the short circuit current is interrupted by the circuit breaker.

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TABLE VIII

WIRE SIZE AND CIRCUIT BREAKER DESIGNATION

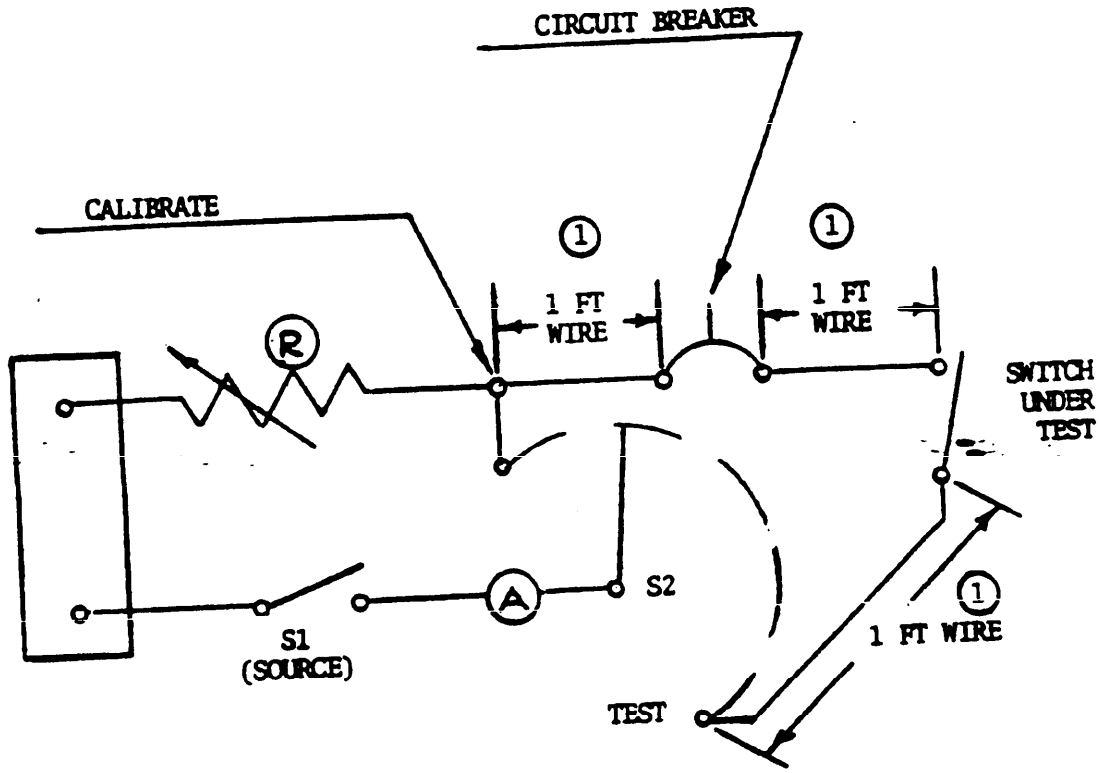
Resistive Rating at Lowest Voltage Rating Amperes	Wire Size MIL-W-5086	Circuit Breaker
Less than 5	AN-20	MIL-F-15160/02 Characteristic A rating, as applicable
5	AN-20	MS-25017-5
7-5	AN-18	MS-25017-7
10	AN-18	MS-25017-10
15	AN-18	MS-25017-15
18	AN-16	MS-25017-20
20	AN-16	MS-25017-20
25	AN-14	MS-25017-25
30	AN-14	MS-25017-30
40	AN-12	MS-25017-50

4.7.21.3 A minimum of two (2) minutes shall elapse between successive closings of the switch under test. The test shall be conducted five (5) times for each pole of all possible closed circuits. No poles or contacts multipole switches shall be connected in series for this test.

4.7.22 Overload Cycling - (See Para 3.8.18) Switches shall be tested to make and break the overload current of a resistive circuit equal to 300 per cent of rated resistive load at the rated dc voltage specified in the detail specifications. The rate of operation shall be 5 or 6 operations per minute for 25 operations. Test circuits shall be so arranged that the current will flow through each pole. No poles or contacts of multipole switches shall be connected in series for this test. The duty cycle shall be a nominal 50 per cent "on" and 50 per cent "off".

4.7.23 Explosion - (See Para 3.8.9) Switches shall be tested

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(1) TOLERANCE = ±0.25 INCH (WIRE SHALL BE CUT TO SPECIFIED LENGTH, STRIPPED AND TERMINATED).
THE TERMINALS SHALL BE IN ACCORDANCE WITH SPECIFICATION MIL-T-7928.
THE CIRCUIT BREAKERS SHALL BE IN ACCORDANCE WITH SPECIFICATION MIL-C-5809 AND TABLE VIII.
THE FUSES SHALL BE IN ACCORDANCE WITH SPECIFICATION MIL-F-15160 AND TABLE VIII

FIGURE II
CIRCUIT DIAGRAM FOR
SHORT CIRCUIT TEST

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4.7.23 (Cont)

in accordance with MIL-STD-202, Method 109. The following detail shall apply.

4.7.23.1 Mechanical and Electrical Load - Switches shall be operated at their maximum rated dc inductive current as specified in the detail specifications.

4.7.24 Electrical Endurance - (See Para 3.8.19) Test loads and circuits shall be so arranged that an independent load of the specified current shall be provided for each pole. The wire used shall be as specified in Table VIII. Load and duty cycle requirements shall be as described in Table IX.

4.7.24.1 Test Procedure - Switches shall be monitored to determine whether any contact has failed to open or close its individual circuit in the proper sequence. Each individual circuit must control the specified load. The monitoring circuit shall not shunt inductive components of inductive loads or switch contacts.

4.7.24.2 High Temperature Endurance - Prior to and following the resistive endurance cycling, perform the following.

4.7.24.2.1 Restrict the switch to one current carrying position. When more than one position of the switch is a conducting position, the positions shall be changed between tests to assure high temperature current carrying capability exists in each position.

4.7.24.2.2 Stabilize the switch temperature at 160°F $^{+5}_{0}$.

4.7.24.2.3 Adjust the load to carry the rated resistive current specified in the detail specification.

4.7.24.2.4 The switch shall carry this current continuously for a minimum of one (1) hour.

4.7.24.2.5 With the switch carrying continuous current (no make or break), transfer

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ELECTRICAL ENDURANCE REQUIREMENTS

Load	Voltage: 28 ± 2 Volts DC	Voltage: 115 ± 5 Volts AC, 400 ± 20 Cycles	Duty Cycle: % On	Duty Cycle: % Off	Duty Cycle: Cycles Per Minute	Temperature	Altitude
Resistive	10 amp	10 amp	50	50	10-12	160°F	Sea Level
Inductive	5 amp	5 amp	50	50	10-12	-65°F	80,000 Ft.
Lamp	2 amp	2 amp	30	70	2-3	160°F	Sea Level
Resistive (High Temp Soak)	10 amp	10 amp	100	--	---	275°F	Sea Level

TABLE IX

LOAD AND DUTY CYCLE REQUIREMENTS

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4.7.24.2.5 (Cont)

the switch to a temperature chamber which has been stabilized at 275°F $^{+5}_{-0}$F. The transfer time shall be 3 minutes maximum. The temperature during transfer time shall be room ambient. The time required for the chamber temperature to stabilize after transfer shall be 3 minutes maximum. The switch shall be left in the chamber for a total time of 15 minutes, including the stabilization time.

4.7.24.2.6 With the switch still carrying continuous current, transfer the switch back to the 160°F chamber and proceed with resistive endurance cycling or to the next step when run after resistive endurance cycling.

4.7.25 Endurance Cycling - Switches shall be subjected to 25,000 operating cycles making and breaking the loads specified in the detail specifications and the following paragraphs. The cycling rate shall be 10-12 cycles per minute. A cycle is defined in para. 4.7.18.4. Test conditions shall be as specified in the detail specifications.

4.7.25.1 Resistive DC - Perform Para.4.7.24.

4.7.25.1.1 Perform dc resistive endurance cycling.

4.7.25.1.2 Perform Para.4.7.24.2.

4.7.25.2 Inductive DC - Inductive dc loads shall use MIL-I-81023A or equivalent inductors so adjusted that the time constant L/R (time required for the current to reach 63.2 per cent of its steady state value) shall not be less than 25 milliseconds.

4.7.25.3 Lamp Load DC - The lamp load shall consist of tungsten lamps having a nominal wattage of 25 to 50 watts, of the voltage specified.

4.7.25.4 Resistive AC

4.7.25.4.1 Perform Para.4.7.24.2.

4.7.25.4.2 Perform ac resistive endurance cycling.

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4.7.25.4.3 Repeat paragraph 4.7.25.2.

4.7.25.5 Inductive AC - Inductive ac load currents shall be rated at .7±0.05 lagging power factor at the voltage specified.

4.7.25.6 Lamp Load AC - The lamp load shall consist of tungsten lamps having a nominal wattage not to exceed 200 watts, of the voltage specified.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging - Perservation and packaging shall be level A or C as specified (See Para 6.2)

5.1.1 Cleaning of Switches - Switches shall be cleaned in accordance with MIL-P-116, Process C-1, unless otherwise specified.

5.1.1.2 Drying - Switches shall be dried in accordance with one or more procedures listed in MIL-P-116.

5.1.1.3 Preservation - None required.

5.1.2 Unit Packaging - Switches shall be packaged in intermediate containers (when specified) conforming to PPP-B-566 or PPP-B-676. Intermediate containers shall be uniform in size and be of minimum cube and tare and shall contain multiples of 5-unit packages not to exceed 100 packages or ten pounds. No intermediate packaging is required when the total quantity shipped is less than 100 units. The unit protection shall be 1A-8 Method of MIL-P-116.

5.2 Packing - Packing shall be level "A", "B" or "C" as specified. (See Para 6.2).

5.2.1 Level "A" - The packaged switches shall be packed in fiberboard containers conforming to PPP-B-636, weather resistant, style optional. In lieu of the closure and waterproof requirements in the appendix of PPP-B-636, closures and waterproofing shall be accomplished by sealing the center seams, ends, edges and manufacturer's joints with waterproof tape, 3 inches wide, conforming to PPP-T-60, Class I. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636. Metal banding shall not be used.

5.2.2 Level "B" - The packaged switches shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style and use requirements optional.

5.2.3 Level "C" - The packaged switches shall be packed

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5.2.3 (Cont)

in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. This pack shall conform to the applicable carrier rules and regulations and may be the suppliers commercial practice when such conforms to the requirements of this level.

5.3 Marking - (See 6.2) In addition to any special marking required by the contract or order, each unit package, intermediate and exterior containers shall be marked in accordance with MIL-STD-129

5.4 General - Exterior containers shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities of identical switches to the greatest extent possible.

6. NOTES

6.1 Intended Use - Unless otherwise specified on the individual Military Specification sheets, switches furnished under this specification are not intended for use in circuits with ratings lower than the intermediate current test as specified on the detail specifications.

6.2 Ordering Data - Procurement documents should specify the following.

(a) Title, number and date of this specification.
(b) Title, number and date of the applicable military standard and the Military Standard part number.
(c) Levels of preservation, packaging, packing and applicable marking (See Sec 5).

6.3 Qualification - With respect to products requiring qualification, awards will be made only for such products that have, prior to the time set for opening bids, been tested and approved for inclusion in the applicable Qualified Products List (QPL) whether or not such products have actually been listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they proposed to offer to the Federal Government tested for qualification, in order that they may be eligible to be

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6.3 (Cont.)

awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from the Naval Air Systems Command, Code AIR-53352A, Washington, D.C. 20360.

6.3 1 Copies of specifications and "Provisions Governing Qualification" may be obtained upon application to Commanding Officer, 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.

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Naval Air Systems Command
Washington, D.C. 20360

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343-219 (A-265)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION
MIL-S-81551(AS) SWITCHES, TOGGLE, HERMETICALLY SEALED, GEN SPEC FOR

ORGANIZATION (Of submitter)	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity)	DATE

DD FORM 1 APR 63 1426

C-0870

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